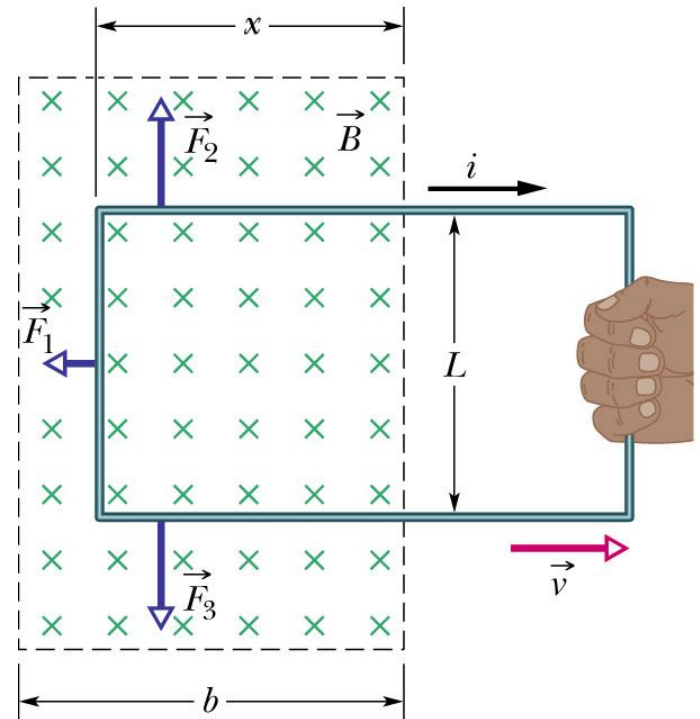
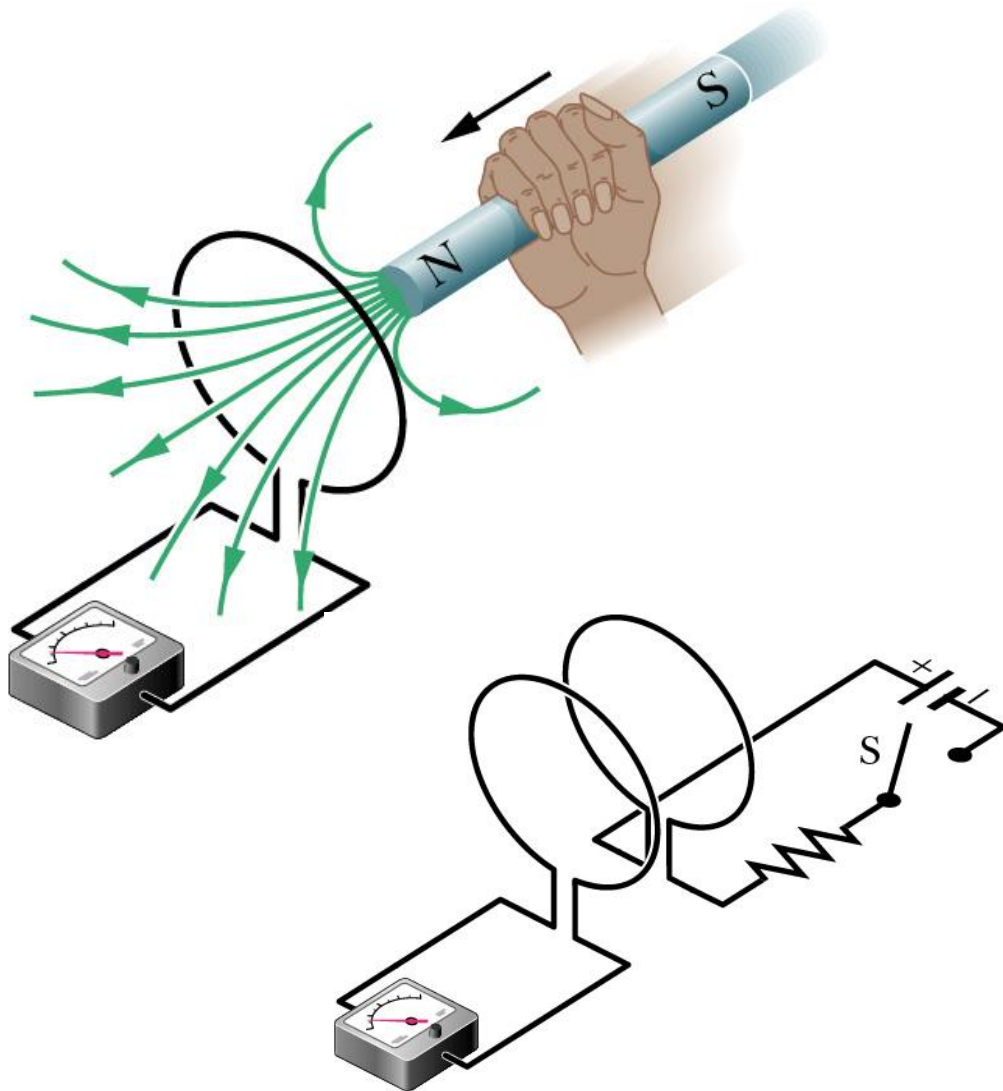


MAXWELL EQUATIONS

Magnetic field as a source of electric field?

Electric field as a source of magnetic field?

Experiments



FARADAY LAW OF INDUCTION

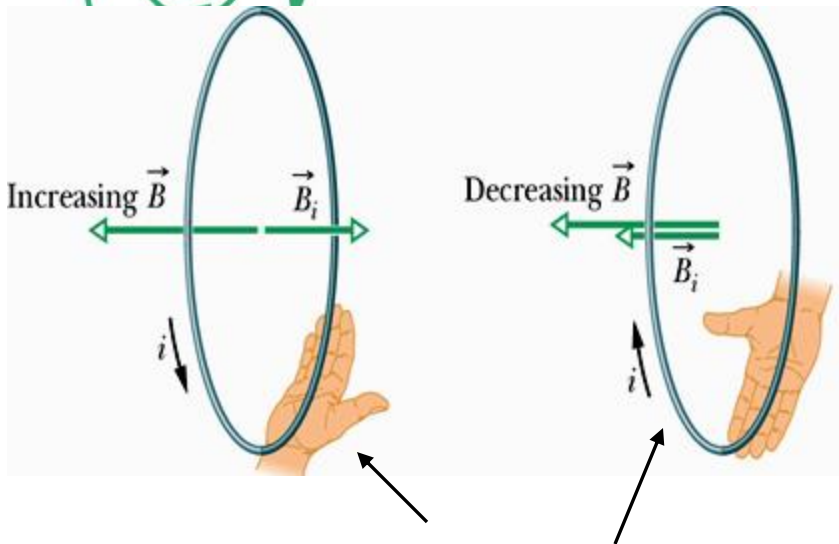
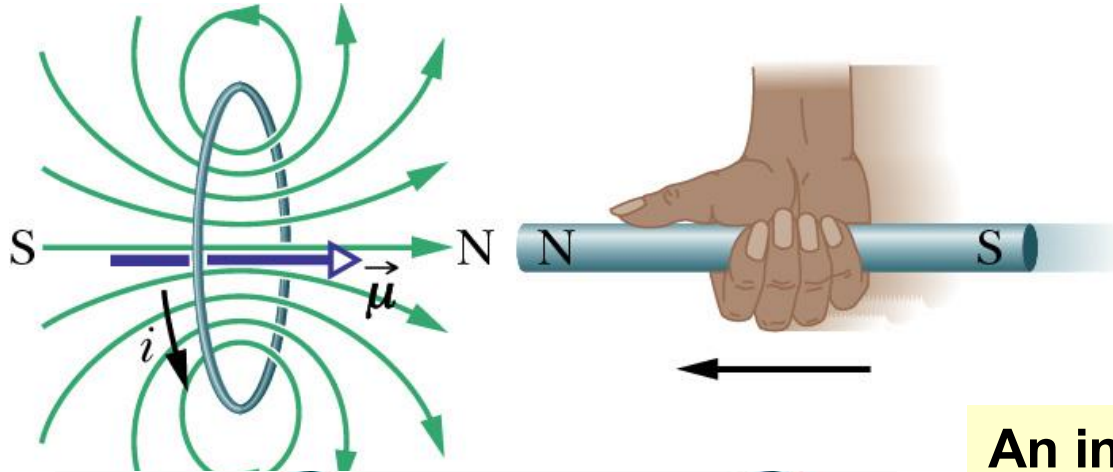
Changing in time the magnetic flux Φ_B induces the electromotive force $\varepsilon = \text{emf}$

$$\varepsilon = - \frac{d\Phi_B}{dt}$$

The biggest rate of change of the magnetic flux, the highest EMF.

The sign „-“ indicates that the result opposes the changes that were its source (Le Chatelier principle, Lenz' law).

LENZ'S LAW



An induced current has a direction such that the magnetic field due to the current opposes the change in the magnetic flux that induces the current.

$\vec{B}_i$ due to the induced current i

Remainder:

Flux of the magnetic field is defined as a surface integral :

$$\Phi_B = \int_S \vec{\mathbf{B}} \circ d\vec{\mathbf{A}}$$

thus there are three ways to induce the electromotive force:

- by a change in the magnetic field B ,
- by a change in the surface S ,
- by a change in the angle between the magnetic field B and the surface vector (rotation of a loop in the magnetic field— current generator)

Remainder: Electromotive force emf of an emf device is the work per unit charge that a device does in moving charge from its low-potential terminal to its high-potential terminal. (work is done by the electric field, magnetic field does no work)

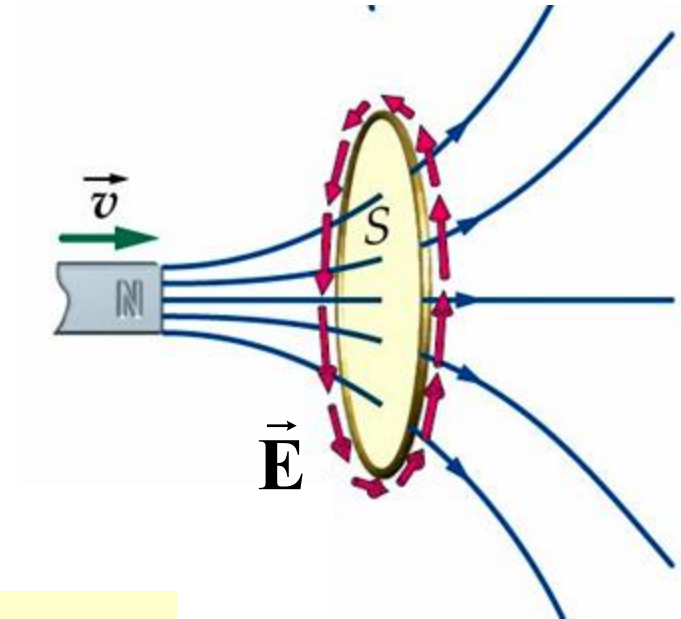
$$\varepsilon = \oint_C \vec{\mathbf{E}} \circ d\vec{\mathbf{l}}$$

Induced emf is due to the work done by the induced electric field, thus

$$\oint_C \vec{\mathbf{E}} \circ d\vec{\mathbf{l}} = -\frac{d\Phi_B}{dt}$$

and

$$\oint_C \vec{\mathbf{E}} \circ d\vec{\mathbf{l}} = -\frac{d}{dt} \int_S \vec{\mathbf{B}} \circ d\vec{\mathbf{A}}$$



This form of Faraday's law is one of the Maxwell's equations in the integral form.

Properties of induced emf

- Induced emf is not localised, (between the terminals of the power source), but is distributed along a whole circuit.
- It can be described as a line integral around a closed path of the induced electric field.
- This integral is non-zero, hence the induced electric field is non-conservative.
- This field has no potential nor equipotential surfaces.
- If a conductor is placed in the region of the induced electric field and the circuit is closed, the induced electric current appears. In the opposite case, we can speak about the electromotive force
- Dissipation of energy takes place when there are the electric charges.

Differential form of Faraday's law

From Stokes' theorem $\oint_C \vec{\mathbf{E}} \circ d\vec{\mathbf{l}} = \int_S (\text{rot } \vec{\mathbf{E}}) \circ d\vec{\mathbf{A}}$

From Faraday's law in the integral form

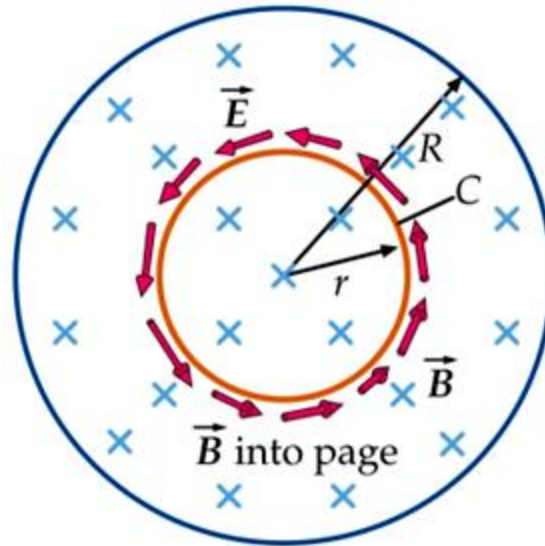
$$\oint_C \vec{\mathbf{E}} \circ d\vec{\mathbf{l}} = - \frac{d}{dt} \int_S \vec{\mathbf{B}} \circ d\vec{\mathbf{A}}$$

$$\text{rot } \vec{\mathbf{E}} = - \frac{\partial \vec{\mathbf{B}}}{\partial t}$$

Time-dependent magnetic field induces an electric field which is rotational, time-dependent and non-conservative.

Example 11.1

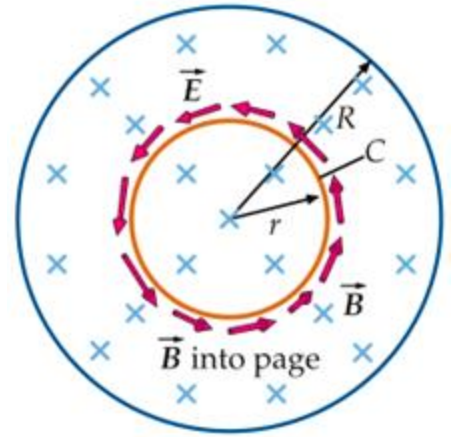
Uniform magnetic field, the vector of which is perpendicular to the plane of the drawing, exists in the cylindrical region of the radius R . Magnitude of the magnetic field changes in time with the rate of dB/dt . Calculate the magnitude of electric field vector induced on the circular cross section shown below. The distance from the center of the circle is r . Consider two cases $r < R$ and $r > R$.



Solution:

From Faraday's law

$$\oint_C \vec{E} \circ d\vec{l} = -\frac{d}{dt} \int_S \vec{B} \circ d\vec{A}$$



we calculate the circulation of $\oint_C \vec{E} \circ d\vec{l} = E 2 \pi r$
the electric field

for $r > R$ the flux of magnetic field

$$\int_S \vec{B} \circ d\vec{A} = B \pi R^2$$

hence

$$E 2 \pi r = -\pi R^2 \frac{dB}{dt}$$

$$E = -\frac{R^2}{2r} \frac{dB}{dt}$$

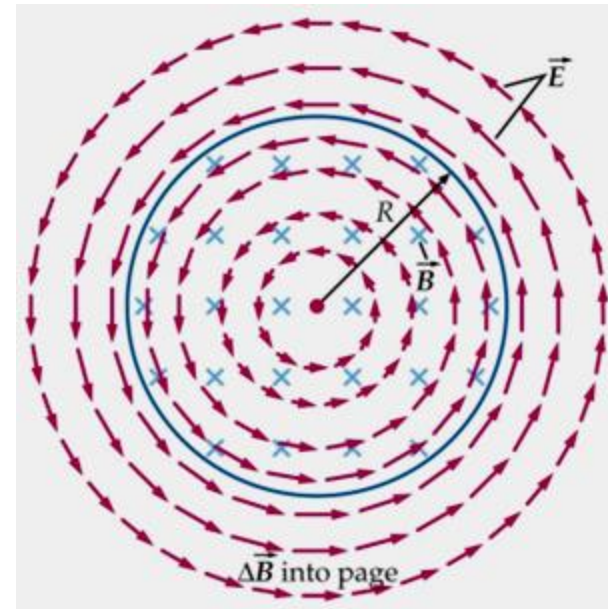
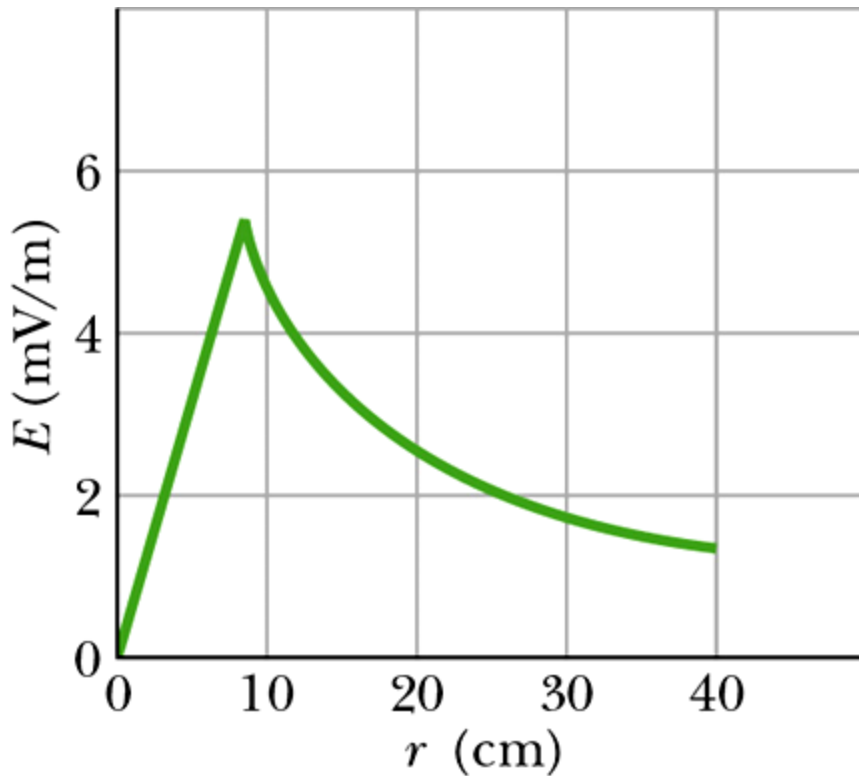
for $r < R$ flux of magnetic field

$$\int_S \vec{B} \cdot d\vec{A} = B \pi r^2$$

thus

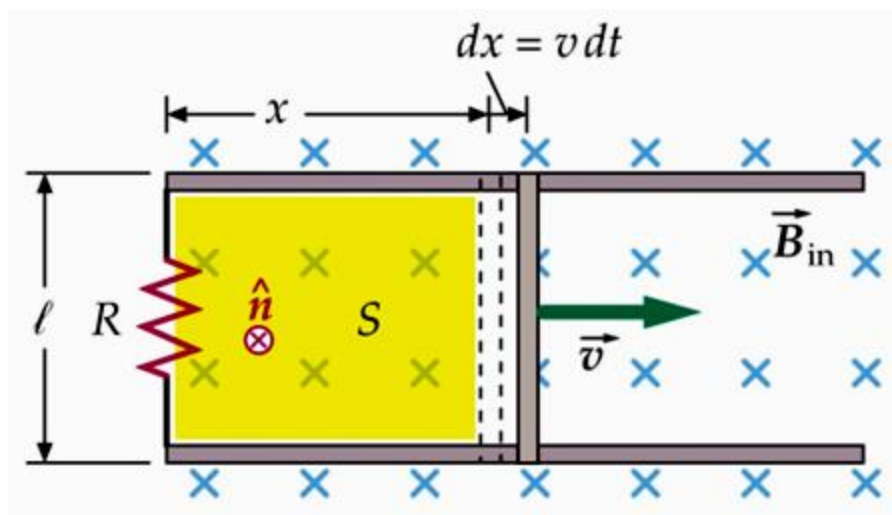
$$E 2 \pi r = - \pi r^2 \frac{dB}{dt}$$

$$E = - \frac{r}{2} \frac{dB}{dt}$$



Homework 11.1

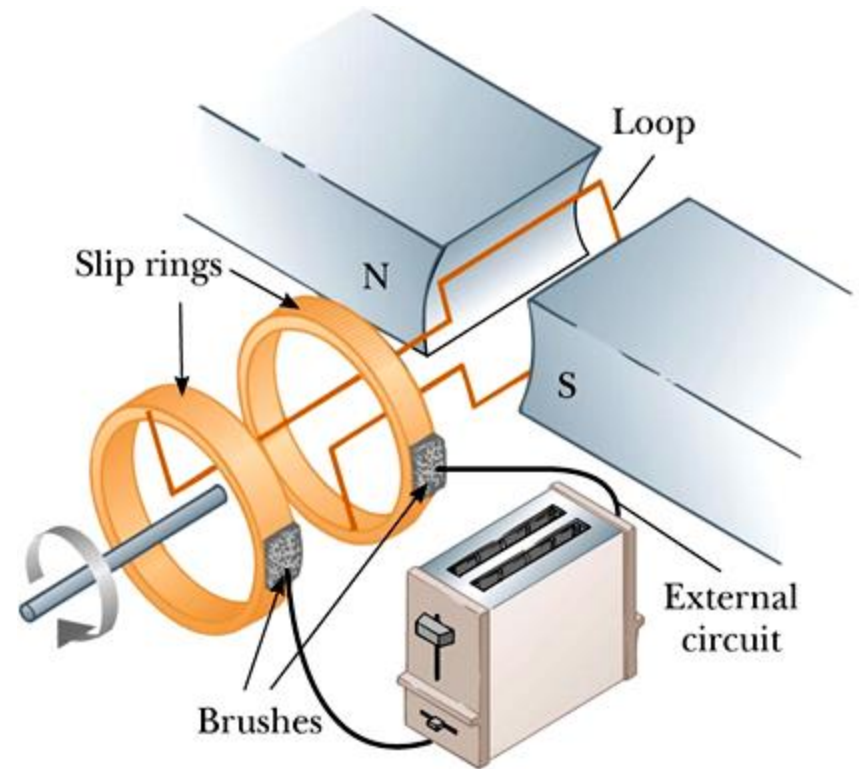
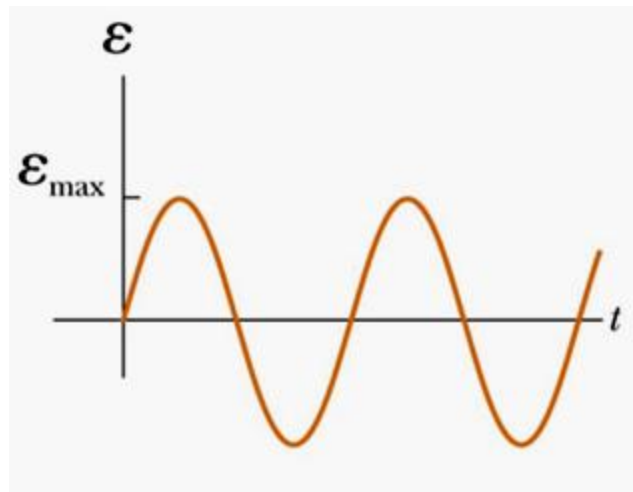
Find the value of emf induced in the electric circuit shown below. Metallic rod is displaced in a time-independent, uniform magnetic field with a constant speed v . Calculate the current induced in this circuit. What is a power dissipated at the resistance R ?



Homework 11.2

- Current generator (ac)

Circular loop of a radius $R=20$ cm contains $N=20$ turns. How fast this loop has to rotate in the magnetic field of $B=0.2$ Wb/m² in order to produce the maximum emf of 160 V?



SELF-INDUCTION

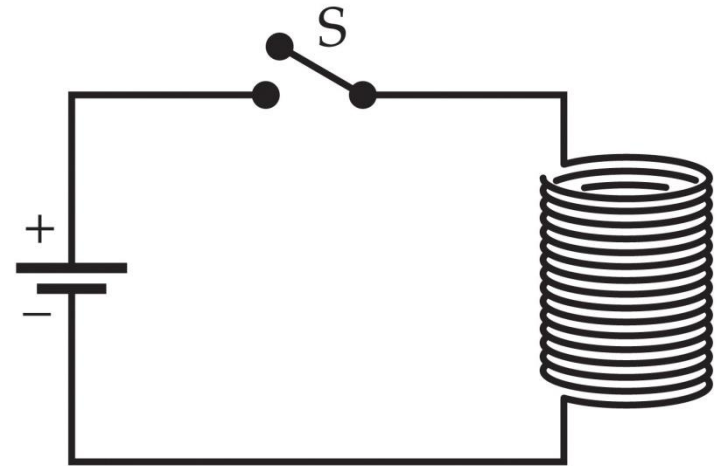
An induced emf appears in any coil in which the current is changing in time. The self-induced emf has the orientation such that it opposes the change in current i (Lenz's law)

Magnetic field inside an ideal solenoid of N turns and length l

$$B = \mu_0 \frac{N}{l} i$$

Magnetic flux through a surface NS (S -cross section area of a single loop)

$$\Phi_B = \mu_0 \frac{N^2}{l} i S$$



$$\Phi_B = Li$$

Unit: 1H (henr) = 1Wb/A

inductance

Emf of self-induction

$$\varepsilon_L = -\frac{d\Phi_B}{dt} = -\mu_o \frac{N^2}{l} S \frac{di}{dt} = -\mu_o n^2 l S \frac{di}{dt}$$

n- number of turns per unit length, l – total length of solenoid, S – cross sectional area

$$\varepsilon_L = -L \frac{di}{dt}$$

In LC circuit one can apply Kirchhoff's laws:



$$L \frac{di}{dt} + \frac{q}{C} = 0 \quad \longrightarrow \quad L \frac{d^2q}{dt^2} + \frac{q}{C} = 0$$

INDUCTANCE

Definition: $L = -\frac{\varepsilon_L}{\frac{di}{dt}}$

Reminder: capacitance C

$$C = \frac{Q}{U}$$

Ideal solenoid

$$L = \mu_o \frac{N^2}{l} S$$

Parallel-plate capacitor

$$C = \varepsilon_o \frac{A}{d}$$

Inductance, similar to the capacitance, depends on the geometric parameters of the device, only. It can be increased by filling the solenoid with ferromagnetic material of the relative magnetic permeability μ .

$$L = \mu_o \mu \frac{N^2}{l} S = \mu_o \mu n^2 l S$$

Homework 11.3

- Prepare the following topics: energy stored in the magnetic field and energy density (HRW, vol.3, 31.10,31.11).
- Consider the analogy to the electric field.

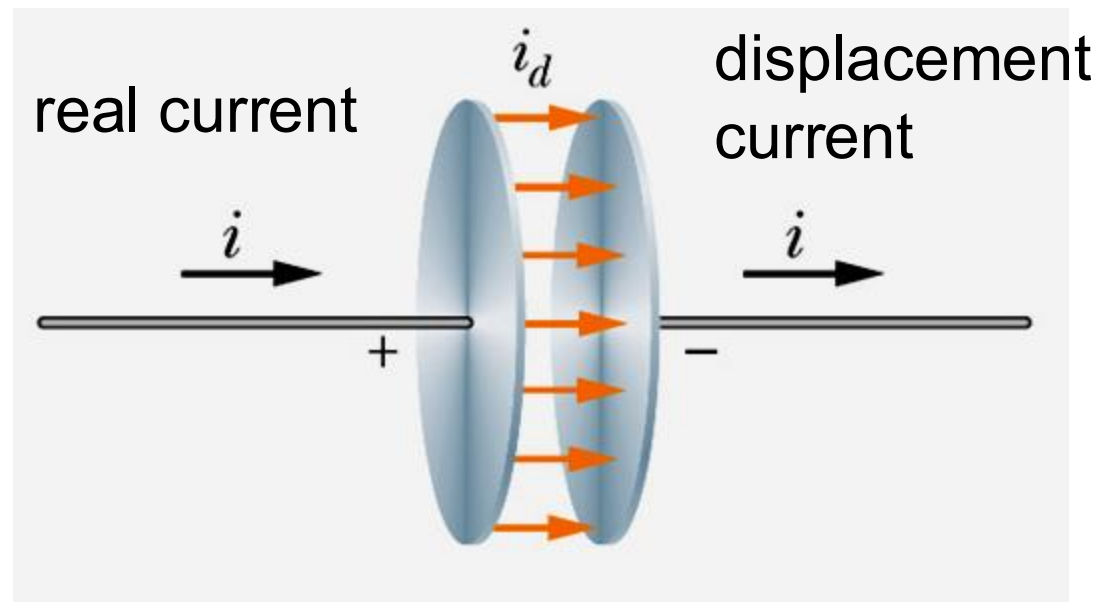
AMPERE'-MAXWELL LAW

- Maxwell extended Ampere law
- As far as the sources of the magnetic field are concerned, not only the real electric current but changing in time flux of the electric field produces the same effect

$$\oint_C \vec{\mathbf{B}} \circ d\vec{\mathbf{l}} = \mu_o \left(i + \underbrace{\varepsilon_o \frac{d\Phi_E}{dt}} \right)$$

Displacement current i_d

- A concept of the displacement current allows to keep the continuity of current in the circuit even when the capacitor is present.



$$i_d = \epsilon_0 \frac{d\Phi_E}{dt}$$

MAXWELL'S EQUATIONS

Law:	Integral form	Differential form
Gauss for electric field	$\oint_S \vec{E} \circ d\vec{A} = \frac{q}{\epsilon_0}$	$\text{div } \vec{E} = \frac{\rho}{\epsilon_0}$
Gauss for magnetic field	$\oint_S \vec{B} \circ d\vec{A} = 0$	$\text{div } \vec{B} = 0$
Ampere-Maxwell's	$\oint_C \vec{B} \circ d\vec{l} = \mu_0 \left(i + \epsilon_0 \frac{d\Phi_E}{dt} \right)$	$\text{rot } \vec{B} = \mu_0 \left(\vec{j} + \epsilon_0 \frac{\partial \vec{E}}{\partial t} \right)$
Faraday's	$\oint_C \vec{E} \circ d\vec{l} = - \frac{d\Phi_B}{dt}$	$\text{rot } \vec{E} = - \frac{\partial \vec{B}}{\partial t}$

ELECTROMAGNETIC WAVE in vacuum

- It is assumed that $j=0$, $\rho=0$
- Maxwell equations:

$$\operatorname{div} \vec{\mathbf{E}} = 0 \quad \longrightarrow \quad \nabla \circ \vec{\mathbf{E}} = 0$$

$$\operatorname{div} \vec{\mathbf{B}} = 0 \quad \longrightarrow \quad \nabla \circ \vec{\mathbf{B}} = 0$$

$$\operatorname{rot} \vec{\mathbf{B}} = \mu_0 \varepsilon_0 \frac{\partial \vec{\mathbf{E}}}{\partial t} \quad \longrightarrow \quad \nabla \times \vec{\mathbf{B}} = \mu_0 \varepsilon_0 \frac{\partial \vec{\mathbf{E}}}{\partial t}$$

$$\operatorname{rot} \vec{\mathbf{E}} = -\frac{\partial \vec{\mathbf{B}}}{\partial t} \quad \longrightarrow \quad \nabla \times \vec{\mathbf{E}} = -\frac{\partial \vec{\mathbf{B}}}{\partial t}$$

Derivation of the EB wave

$$\nabla \times (\nabla \times \vec{\mathbf{E}}) = -\nabla \times \frac{\partial \vec{\mathbf{B}}}{\partial t} = -\frac{\partial}{\partial t} \nabla \times \vec{\mathbf{B}}$$

but $\nabla \times \vec{\mathbf{B}} = \mu_0 \epsilon_0 \frac{\partial \vec{\mathbf{E}}}{\partial t}$

hence:

$$\nabla \times (\nabla \times \vec{\mathbf{E}}) = -\mu_0 \epsilon_0 \frac{\partial^2 \vec{\mathbf{E}}}{\partial t^2} \quad (1)$$

Using the vectorial identity: $\vec{\mathbf{a}} \times (\vec{\mathbf{b}} \times \vec{\mathbf{c}}) = (\vec{\mathbf{a}} \circ \vec{\mathbf{c}}) \vec{\mathbf{b}} - (\vec{\mathbf{a}} \circ \vec{\mathbf{b}}) \vec{\mathbf{c}}$

$$\nabla \times (\nabla \times \vec{\mathbf{E}}) = (\underbrace{\nabla \circ \vec{\mathbf{E}}}_0) \nabla - \nabla^2 \vec{\mathbf{E}} \quad (2)$$

Combining (1)
and (2) we get:

$$\nabla^2 \vec{\mathbf{E}} = \mu_0 \epsilon_0 \frac{\partial^2 \vec{\mathbf{E}}}{\partial t^2}$$

General differential equation:

$$\nabla^2 \Psi(\vec{\mathbf{r}}, t) = \frac{1}{v^2} \frac{\partial^2 \Psi}{\partial t^2}$$

VELOCITY OF ELECTROMAGNETIC WAVE IN VACUUM

Wave equations:

$$\begin{array}{ll}\text{For a magnetic field} & \nabla^2 \vec{\mathbf{B}} = \mu_0 \epsilon_0 \frac{\partial^2 \vec{\mathbf{B}}}{\partial t^2} \\ \text{For an electric field} & \nabla^2 \vec{\mathbf{E}} = \mu_0 \epsilon_0 \frac{\partial^2 \vec{\mathbf{E}}}{\partial t^2}\end{array}$$

Perturbation ψ is the vector of the electric field $\mathbf{E}$ or magnetic field $\mathbf{B}$ and the phase velocity v is determined by the universal constants, only:

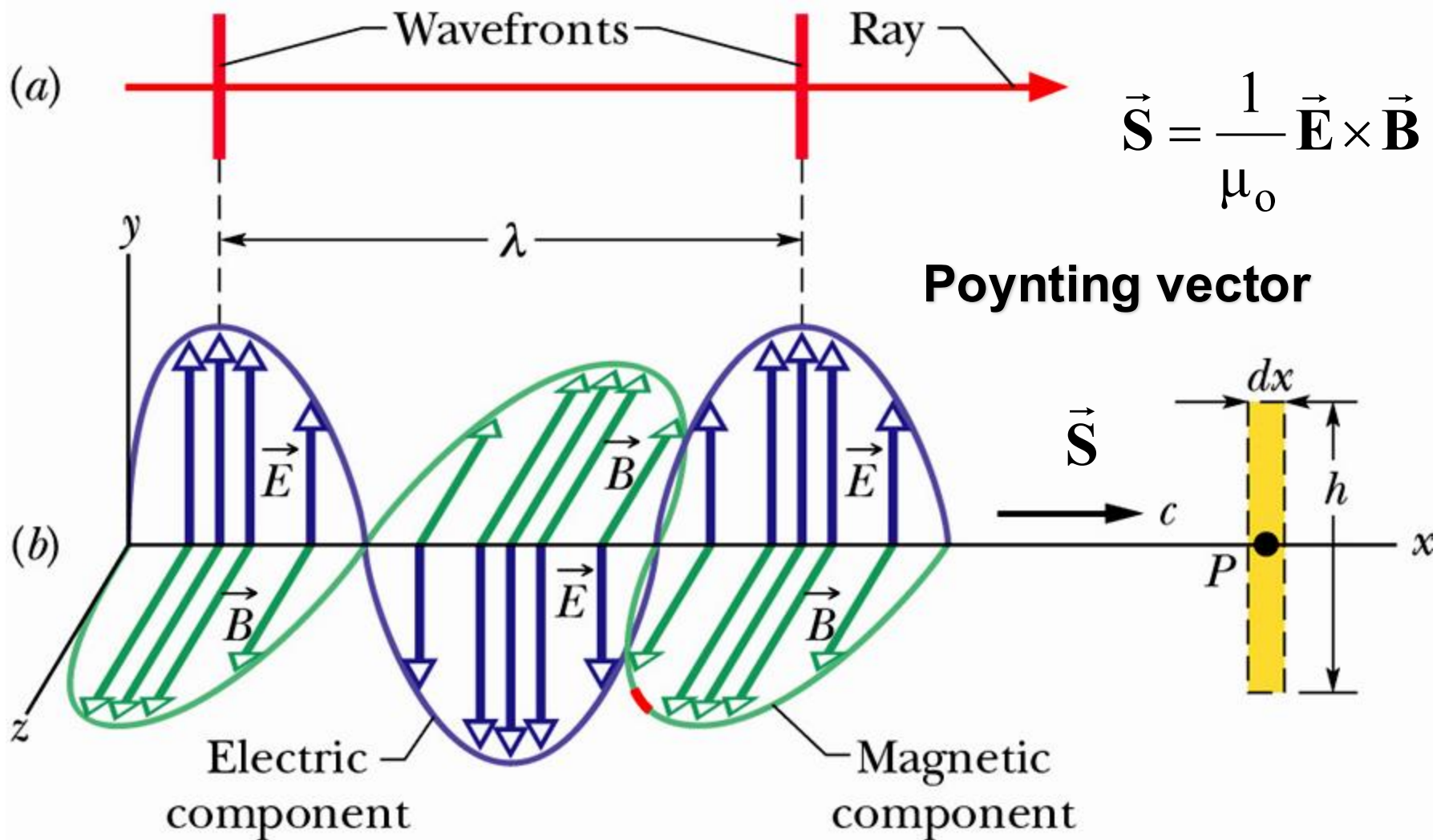
$$v = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = c$$

velocity of EB wave (light velocity) in vacuum can be calculated theoretically ($c \approx 3 \cdot 10^8$ m/s)

Homework 11.4

- Prepare a topic: „Experimental methods of determination of light velocity in vacuum”

Propagation of electromagnetic wave



Summary

- Time-dependent magnetic field is a source of the electric field (Faraday's law)
- Induced electric field is non-conservative, rotational with non-zero curl
- Time-dependent electric field is a source of magnetic field (Maxwell's modification of Ampère law)
- Maxwell's equations in vacuum are symmetric for both fields
- Maxwell's equations predict the existence of electromagnetic wave, propagating in vacuum with the phase velocity c